

D.W& D.M Important questions Answers

1. Select one data pre-processing technique and explain how we improve data quality in DM

Ans: Data Cleaning as a Data Pre-processing Technique: Data cleaning is a data pre-processing technique used to improve data quality by removing errors, missing values, and inconsistent information from a dataset. In data mining (DM), raw data is often incomplete or noisy, which can lead to wrong analysis. Data cleaning corrects these problems by filling missing values, removing duplicates, and fixing incorrect entries. This process makes the dataset accurate, consistent, and reliable, which improves the performance of data mining algorithms.

Example: If a student database contains missing marks or duplicate records, data cleaning fills the missing values and removes duplicates so the analysis gives correct results.

2. How data integration contributes to data mining by combining data from multiple sources. Provide an example.

Ans: Data integration is the process of combining data from multiple sources into a single unified dataset. It improves data mining by providing a complete and consistent view of information. When data comes from different databases, formats, or departments, it may contain conflicts or duplication. Data integration resolves these differences and merges the data into a clean structure. This helps data mining systems discover better patterns and more accurate knowledge because the dataset becomes richer and more reliable.

Example: A company combines customer data from sales, website activity, and social media into one database. After integration, data mining can analyze customer behavior more accurately.

3. Explain the process of Data Reduction & its importance in simplifying datasets for efficient data mining.

Ans: Data reduction is the process of reducing the size of a dataset while keeping the important information intact. It simplifies large datasets by removing unnecessary or repeated data, selecting important features, or compressing data. In data mining, very large datasets increase processing time and complexity. Data reduction improves efficiency by making the dataset smaller, faster to analyze, and easier to understand without losing meaningful patterns. This leads to quicker computation and better performance of data mining algorithms.

Example: In a sales dataset with hundreds of features, only important attributes like price, quantity, and date are selected. This reduced dataset speeds up analysis while still giving useful insights.

4. Discuss 2 different methods of Data discretization used in Data Mining.

Ans: Data discretization is the process of converting continuous data into a small number of intervals or categories. It simplifies numeric data so that data mining algorithms can work faster and more efficiently.

1. Equal Width Discretization

In this method, the range of data is divided into equal-sized intervals. Each interval has the same width, regardless of how many data points fall inside it. This method is simple and easy to implement but may not distribute data evenly.

2. Equal Frequency Discretization

In this method, data is divided so that each interval contains approximately the same number of data points. This creates balanced groups and is useful when data is unevenly distributed.

Example: If student marks from 0–100 are discretized, equal width may divide them into 0–25, 26–50, 51–75, 76–100, while equal frequency divides marks so each group contains the same number of students.

5. What is the significance of Data Transformation in DM.

Ans: Data transformation converts raw data into a suitable and consistent format for analysis. It improves data quality by normalizing values, removing noise, and organizing data properly. This makes data mining faster, more accurate, and efficient because algorithms work better on clean and structured data.

Example: Converting income values from different currencies into one standard currency for proper analysis.

6. Explain the significance of Data discretization.

Ans: Data discretization converts continuous data into categories or intervals, which simplifies large datasets and makes them easier to analyze. It reduces complexity, speeds up data mining algorithms, and improves understanding of patterns by grouping similar values together.

Example: Student marks can be grouped as low, medium, and high instead of using exact numbers.

7. Describe features of DM & its applications.

Features of Data Mining (DM)

1. Extracts hidden patterns from large datasets
2. Handles large and complex data efficiently
3. Supports decision making
4. Uses statistical and AI techniques
5. Improves prediction and classification

Applications of Data Mining

1. Business – customer behavior analysis
2. Healthcare – disease prediction
3. Banking – fraud detection
4. Education – student performance analysis
5. Marketing – targeted advertising

8. Explain the concept of Association Rule Mining.

Ans: Association rule mining is a data mining technique used to discover relationships and patterns between items in a large dataset. It identifies which items frequently occur together and forms rules to predict item combinations. It is mainly used in market basket analysis to understand customer buying behavior.

Example: In a supermarket, if customers often buy bread and butter together, the system creates a rule showing this association to improve product placement.

9. Explain the knowledge discovery process in DM.

Ans: Knowledge discovery in data mining is a step-by-step process used to extract useful information from large datasets. It begins with selecting relevant data, then cleaning and transforming it to remove errors and inconsistencies. After that, data mining techniques are applied to find patterns and relationships. Finally, the discovered knowledge is evaluated and presented in an understandable form for decision making.

Example: A company collects sales data, cleans and organizes it, mines patterns of customer behavior, and uses the results to improve marketing strategies.

10. What are the key steps involved in concept hierarchy generation?

Ans: Concept hierarchy generation organizes data from detailed levels to higher abstract levels to simplify analysis. The main steps include identifying attributes, grouping related values, defining levels of abstraction, and arranging them into a hierarchy. This process helps summarize data and makes data mining more efficient by reducing detail.

Example: Location data can be organized as city → state → country to create a hierarchy.

11. Describe the role of Regression analysis in DM.

1. **Prediction:** Estimates future numeric values based on past data.
2. **Relationship modeling:** Shows how variables are connected and influence each other.
3. **Trend analysis:** Identifies patterns and trends over time.
4. **Decision support:** Helps organizations plan using predicted results.
5. **Data summarization:** Represents complex data with a mathematical model.

Example: A company predicts next month's sales using past sales records.

12. Define data warehouses & their types in detail.

Ans: A data warehouse is a centralized system that stores large amounts of integrated data collected from different sources. It is designed for analysis and decision making rather than daily operations. The data inside a warehouse is cleaned, organized, and stored in a structured way so that it can be easily accessed for reporting and data mining.

Types of Data Warehouses

1. **Enterprise Data Warehouse (EDW):** A large centralized warehouse that stores data for the entire organization and supports company-wide analysis.
2. **Data Mart:** A smaller warehouse focused on a specific department such as sales, finance, or marketing.
3. **Virtual Warehouse:** A system that provides a virtual view of operational databases without storing all data physically.

13. What are the steps to design a Data Warehouse?

1. **Requirement analysis:** Identify business needs and data requirements.
2. **Data source identification:** Select internal and external data sources.
3. **Data modeling:** Design schema and structure of the warehouse.
4. **ETL process design:** Extract, transform, and load data into the warehouse.
5. **Storage design:** Organize data for efficient retrieval.
6. **Testing and deployment:** Verify accuracy and make the warehouse operational.

Example: A retail company designs a warehouse to analyze sales by collecting data from billing, inventory, and customer systems.

14. List the characteristics of DW in detail.

Ans:

1. **Subject-oriented:** Data is organized around major subjects like sales, customers, or products for better analysis.
2. **Integrated:** Data from multiple sources is combined into a consistent format.
3. **Time-variant:** Data is stored with time information to track historical trends.
4. **Non-volatile:** Data is stable and not frequently changed once stored.
5. **Large storage capacity:** Designed to handle huge volumes of data.
6. **Supports decision making:** Optimized for analysis, reporting, and data mining.

15. Which one is faster multidimensional OLAP & What is relational OLAP?

Ans: Multidimensional OLAP (MOLAP) is faster because it stores data in a special multidimensional cube where summaries are already calculated in advance. When a user asks a query, the system does not need to compute again; it simply reads the stored result. This reduces processing time and gives very fast performance, especially for repeated analytical queries.

Relational OLAP (ROLAP) stores data in normal relational database tables instead of cubes. When a query is made, the system calculates the result using SQL operations at that moment. Because calculations happen during query time, it is slower than MOLAP, but it can handle very large datasets and is more flexible.

16. Name some popular DW tools & technologies.

- 1. **Informatica:** Used for data integration and ETL processing.
- 2. **Microsoft SQL Server (SSIS):** Supports data warehousing and transformation.
- 3. **Oracle Data Warehouse:** Enterprise-level warehouse solution.
- 4. **Amazon Redshift:** Cloud-based data warehouse service.
- 5. **Snowflake:** Modern cloud data warehouse platform.

17. Explain the concept of ETL in DW.

Ans: ETL stands for Extract, Transform, and Load, which is the main process used to move data into a data warehouse. In the extract step, data is collected from different sources such as databases or files. In the transform step, the data is cleaned, formatted, and organized into a consistent structure. In the load step, the processed data is stored into the warehouse for analysis. ETL ensures that the warehouse contains accurate and high-quality data.

Example: A company extracts sales data from multiple branch systems, transforms it into a common format, and loads it into a central warehouse for reporting.

18. Describe the role of Data Mart in the DW environment.

Ans: A data mart plays an important role in a data warehouse by providing a focused and simplified view of data for a specific department or user group. Instead of accessing the full warehouse, departments use a data mart that contains only relevant information. This reduces complexity, improves query speed, and supports faster decision making. Data marts also increase security because each department accesses only its own data.

Example: The finance department uses a finance data mart containing budget and expense data to prepare reports quickly.

19. Discuss various techniques of Data Summarization in DW.

Ans: Data summarization reduces detailed data into a compact form so it becomes easier to analyze and understand.

- 1. **Aggregation:** Combines detailed data into totals, averages, or counts for quick analysis.
- 2. **Generalization:** Replaces low-level data with higher-level concepts using hierarchies.
- 3. **Data Cubes:** Pre-computed summaries stored in multidimensional form.
- 4. **Statistical summarization:** Uses measures like mean, variance, and percentage.
- 5. **Dimensional reduction:** Removes unnecessary attributes while keeping important information.

Example: Daily sales data is summarized into monthly or yearly totals for management reports.

20. Analyze & Design DW Schema for an extensive complex dataset.

Ans: Designing a data warehouse schema for a large and complex dataset involves organizing data into a structure that supports fast analysis. The most common design is the **star schema** or **snowflake schema**, where a central fact table is connected to multiple dimension tables. The fact table stores measurable data like sales amount or quantity, while dimension tables store descriptive data such as time, customer, and product. This design reduces complexity, improves query performance, and makes analysis easier. Key steps include identifying business processes, selecting facts and dimensions, defining relationships, and optimizing storage. A good schema ensures consistency, scalability, and efficient reporting.

Example: In a retail warehouse, a sales fact table connects to dimensions like product, customer, store, and time to analyze performance.

21. Differentiate b/w Fact Table Vs Dimension Table.

Feature	Fact Table	Dimension Table
Purpose	Stores measurable data	Stores descriptive information
Data type	Numeric values (facts)	Text or categorical data
Size	Large and detailed	Smaller compared to fact table
Keys	Contains foreign keys	Contains primary keys
Role	Used for calculations and analysis	Provides context to facts

MCQ

1. What is the term in dimension modeling referred to in data warehouse?

Ans: Organizing data for analytical purposes.

2. Which of the following is not a component of DW architecture?

Ans: Data visualization and tools.

3. What is the purpose of a data mart in a DW?

Ans: Provides a subset of data for a specific business unit.

4. Which term refers to the process of transferring data to a format for storage in DW?

Ans: Data transformation.

5. Which characteristic of Data Warehouse compared to operational database?

Ans: Subject-oriented.

6. What is the primary source of data for a data warehouse?

Ans: OLTP system.

7. Purpose of metadata repository in data warehouse?

Ans: Describes and organizes data warehouse.

8. Define key granularity in data warehouse.

Ans: Level of detail in data.

9. Purpose of data replication in data warehouse?

Ans: Increase data availability.

10. What role does data cleaning play in DW?

Ans: Removes inconsistencies from data.

11. Purpose of indexing?

Ans: Enhancing data retrieval performance.

12. Benefit of using DW for reporting and analysis?

Ans: Improves decision-making process.

13. Which OLAP operator views data from different angles/perspectives?

Ans: Dice.

14. Data mart that incorporates data mining tools to extract a set of data?

Ans: Dependent data mart.

15. Components of data warehouse?

Ans: Component key.

16. Define data integration.

Ans: Combining data from different sources to provide a unified view.

17. Which OLAP operation allows users to view data in more detailed dimension?

Ans: Drill down.

18. Roll-up primarily uses?

Ans: Relational database.

19. Which OLAP operation summarizes data along dimensions?

Ans: Roll up.

20. Roll-up concept hierarchy in data mining?

Ans: Multi-level pattern discovery.

21. Subject-oriented, integrated, non-volatile collection of data used to support management decisions is called?

Ans: Data warehouse.

22. Exemplifying DW is called what warehouse?

Ans: Read-only warehouse.

23. Important aspect of data warehouse is that?

Ans: Data warehouse is read-only.

24. Time horizon in data warehouse is usually?

Ans: Simultaneous.

25. Data is stored, retrieved, updated in?

Ans: OLTP.

26. Correct definition of knowledge discovery process?

Ans: Process of extracting useful knowledge from data.

27. Primarily concerned with hierarchy generation in data mining?

Ans: Creates hierarchy structure for categorical data.

28. DW contains what type of data not found in operational environment?

Ans: Summary data.

29. Source of all data in DW is?

Ans: Operational environment.

30. ACID features are used in?

Ans: Online processing environment.

31. Detached data in a single fact table is known as?

Ans: Atomic data.

32. Why is Apriori algorithm selected for mining association rules in large database?

Ans: Efficient pruning of search space.

33. Key property in Apriori algorithm?

Ans: Downward closure property.